

# Visual Library Around One Wave Vocabulary

## Core Idea

`p5.waves` is not only a tool for making a wave move. It can become a source vocabulary for many small visual decisions. The goal of this project is to build a visual library of small, reusable graphic elements that can be copied, adapted, and pasted into larger projects.

The project does not need to start from one fixed content idea, such as a loom, scanner, archive, or instrument. A metaphor can help when it gives the system more focus, but it is not the main goal. The main goal is to collect useful visual modules that show how wave behavior can become design behavior.

The important shift is this:

The library is not the effect. The library is the vocabulary.

A single wave result can be modest: a number between `-1` and `1`, a pulsing value, a changing name, a shifted target, or a repeated curve. The depth appears when those small results are assigned practical graphic jobs.

For example, the same wave value can move a line, choose a word, open spacing, activate a grid cell, bend a title, change a status indicator, or decide when a visual layer appears.

## Expected Result

The expected result is a website that functions as a visual library of small graphic elements. Each element demonstrates one possible use of `p5.waves` and is presented in a way that makes it easy to understand, copy, adapt, and reuse in larger projects.

The site can take many forms: a catalogue, archive, specimen page, grid, collection, gallery, dashboard, or toolbox. A dashboard is only one possible form, not the main goal. The main goal is the library itself: a growing collection of reusable wave-driven visual elements.

Each element should include:

- **A visual example:** a small live graphic, animation, pattern, interface part, or typographic behavior.
- **The code:** the minimal code needed to understand or reuse the element.
- **A short text:** a simple explanation of what the element does and which wave behavior drives it.

The site becomes both a visual catalogue and a practical toolbox. It should help a maker browse, compare, copy, remix, and combine wave-driven design ideas.

## Why Mini Applications Matter

The strength of `p5.waves` is not that one call creates a finished artwork. Its strength is that the same small call can be reused in many contexts.

A wave value can:

- move a label
- pick a color
- gate a word
- bend a line
- select a pattern
- change a grid threshold
- animate a progress state
- drive a scanner
- open and close spacing
- choose between visual voices
- make a title breathe
- turn a row or column on
- reveal a system status
- create a printable texture
- distort a grid
- control a mask
- shift a border
- trigger a layout change
- create a loading state
- make an interface feel alive

Each use is small, but each use proves that wave behavior can become design behavior. The library becomes interesting when it is treated as a set of micro-decisions, not as one large animation trick.

## What The Visual Library Is

The visual library is a curated collection of small graphic modules, presented inside a website. It is not a finished artwork with one single story. It is also not just a list of unrelated demos. It sits between both.

Each module should answer this question:

What small graphic job does this wave behavior perform?

A strong module has:

- a clear visual behavior
- a simple use case
- readable code
- a short explanation
- a reason to be reused

The full dashboard should make it easy to see how many different design tasks can grow from the same wave vocabulary.

## Site Structure

A useful site could contain cards or entries. Each card represents one graphic element.

Each card could contain:

- **Name:** a clear title for the element.
- **Role:** what job the wave performs.
- **Preview:** a small running visual example.
- **Code:** the core snippet or full mini sketch.
- **Notes:** a short explanation of how and why it works.
- **Tags:** motion, type, grid, color, texture, state, interface, export, pattern, layout.

The site itself can also be wave-driven. For example, `p5.waves` could influence entry order, preview motion, active filters, hover behavior, transitions, or visual grouping.

## From Function To Reusable Element

The path from simple wave result to reusable graphic element can be described in five steps.

### 1. Primitive

Start with one technical result.

Examples:

- `Waves.wave(x, options)` returns a curve value.
- `sampler.sample(i, t)` returns a changing value.
- `sampler.waveName` exposes the current wave.
- `sampler.targetName` exposes where the shift is going.
- `sampler.mix` exposes the transition state.

At this stage the result is not yet a design element. It is only raw behavior.

### 2. Role

Give the primitive a graphic job.

Examples:

- a value becomes tension
- a pulse becomes permission
- a mix value becomes progress
- a wave name becomes a label
- row and column values become a pattern decision
- a sampled value becomes spacing
- a threshold becomes a visibility rule

This is where the design starts.

### 3. Element

Wrap the role in a small visual application.

Examples:

- a pattern card
- a status strip
- a type specimen
- a ribbon field
- a threshold grid
- a signal meter
- a moving annotation layer
- an animated divider
- a breathing title
- a flickering label
- a wave-driven button state

Each element should make visual sense on its own.

#### **4. Library Entry**

Document the element so it can be reused.

A good library entry explains:

- what the element is
- what wave behavior it uses
- which part of the code matters
- where it could be reused
- how it could be varied

This turns a sketch detail into a reusable design part.

#### **5. Site Entry**

Place the element inside the larger visual library site.

The site collects all entries and makes the library visible. It should feel like a living catalogue of graphic possibilities.

The result does not need to prove one content metaphor. It needs to prove that one wave vocabulary can produce many useful visual decisions.

### **Criteria For A Strong Library Element**

A library element is strong when:

- the graphic behavior is easy to understand
- the wave has a clear role
- the code is small enough to reuse
- the element can work inside a larger project
- the result feels designed, not random
- the visual idea can generate variations
- the explanation is short and useful

## Criteria For A Strong Site

The site is strong when:

- it contains many different types of graphic jobs
- each element has a preview, code, and short text
- the entries are easy to browse
- the visual system feels coherent
- the library feels useful for future projects
- the code can be copied without needing too much context
- the whole site shows `p5.waves` as a design language rather than a plugin

## Possible Categories

The dashboard can organize elements by category.

### Motion

Waves control movement, position, rotation, speed, easing, delay, or repetition.

Possible elements:

- breathing title
- drifting label
- scanning line
- oscillating marker
- moving annotation
- wave-following cursor

### Type

Waves control typographic behavior.

Possible elements:

- shifting letter spacing
- flickering word gate
- animated headline
- wave-driven word picker
- unstable caption
- changing type label

### Grid

Waves control cells, rows, columns, thresholds, and pattern decisions.

Possible elements:

- active cell grid
- threshold field

- row scanner
- column pulse
- woven block pattern
- density map

## **Color**

Waves control color choice, intensity, contrast, opacity, or palette shifts.

Possible elements:

- pulsing swatch
- changing background
- wave-based color picker
- opacity rhythm
- contrast meter
- signal palette

## **Texture**

Waves create printable or screen-based textures.

Possible elements:

- halftone field
- noisy stripe pattern
- pressure texture
- wave grain
- distorted hatch
- exportable pattern tile

## **Interface**

Waves make interface elements feel alive.

Possible elements:

- loading state
- progress meter
- status indicator
- active tab signal
- hover pulse
- notification rhythm

## **State**

Waves control changes between modes or conditions.

Possible elements:

- transition meter

- mix readout
- active wave label
- target wave label
- mode switcher
- signal lock indicator

## Layout

Waves influence spacing, alignment, grouping, and hierarchy.

Possible elements:

- expanding margin
- shifting column width
- breathing card grid
- rhythmic divider
- adaptive spacing system
- animated section marker

## Optional Metaphors

A fixed metaphor is not required. However, a metaphor can still help if the site starts to feel too loose.

Possible metaphors:

- signal room
- visual workshop
- design laboratory
- wave archive
- motion specimen cabinet
- interface machine
- graphic toolbox

The metaphor should support the library. It should not trap the project inside one theme.

## Working Principle

For every future element, start with this question:

What tiny graphic job can this wave behavior perform?

Then turn that job into a reusable card with a preview, code, and short explanation.

The final result should not feel like one finished artwork or a page of random examples. It should feel like a living visual library: a website full of small, useful, wave-driven graphic elements that can be copied, adapted, and combined inside larger projects.